DBMS – Sample Study Notes

CampusAI demonstration material. These are sample notes; verify and expand them using your course syllabus.

What is DBMS?

- A Database Management System (DBMS) stores, organizes, retrieves, and manages data.
- Examples include MySQL, Oracle Database, PostgreSQL, and SQL Server.

Advantages of DBMS

- Reduces unnecessary duplicate data.
- Helps keep data consistent and secure.
- Supports efficient data retrieval, backup, and recovery.

Keys in DBMS

- Primary key: uniquely identifies each row.
- Foreign key: connects tables by referring to a key in another table.
- Candidate key: a field or group of fields that can uniquely identify a row.

SQL

- SQL means Structured Query Language.
- SELECT retrieves data; INSERT adds data; UPDATE changes data; DELETE removes data.
- Example: SELECT name FROM Student;

Normalization

- Normalization organizes tables to reduce duplication and update problems.
- 1NF: each cell contains a single value.
- 2NF: 1NF and no partial dependency on a composite key.
- 3NF: 2NF and no transitive dependency on a key.